

How Much Does Input Curation Alone Improve 3D Gaussian Splatting from Casual Video?

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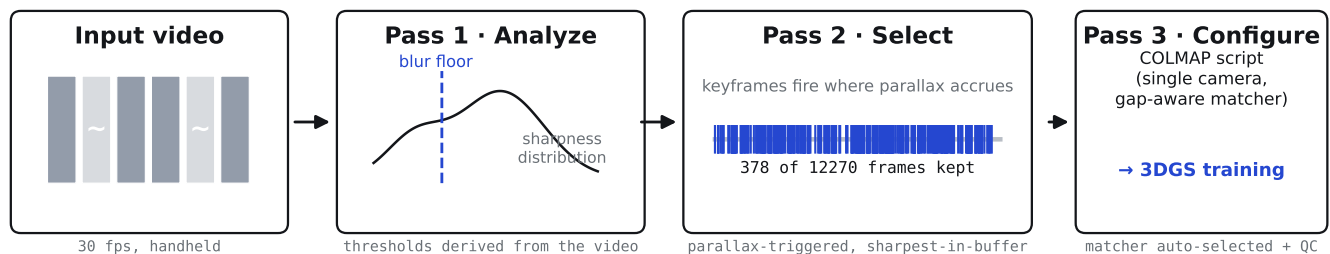


Figure 1: Pipeline overview: an analysis pass derives all thresholds from the video itself; keyframes are triggered by tracked parallax (sharpest frame in a short buffer kept); the output is QC'd and a ready-to-run COLMAP configuration is emitted. Only this input stage differs between arms.

Abstract

The standard recipe for reconstructing a 3D Gaussian Splatting (3DGS) scene from a casually captured video starts with fixed-interval frame extraction — a step widely believed to admit motion blur, redundancy, and uneven baselines into the pipeline. We quantify what fixing *only* this step actually buys, evaluating a fully automatic parallax-and-sharpness-based selector under a strictly controlled protocol (identical frame budget, COLMAP settings, and 3DGS training). The answer: no improvement — on average curation *lowers* held-out PSNR by 0.70 dB across 10 real scenes (bootstrap CI excludes zero) *even though it improves SfM statistics*. The reason is a strong per-view correlation: quality differences track the distance to the nearest training view (Spearman $\rho = -0.63$ over 752 views) — curation wins where its spacing is locally denser, loses where parallax-triggering thins coverage. Under synthetic exposure blur the mean PSNR deficit shrinks monotonically (-0.93 to -0.03 dB), approaching near parity only at the heaviest degradation;

on ordinary captures, temporal coverage — not frame sharpness — predicts quality.

CCS Concepts

• **Computing methodologies** → **Shape modeling**; *Image manipulation*.

Keywords

3D Gaussian Splatting, keyframe selection, input curation, COLMAP, novel view synthesis

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1 Introduction

Video is the most natural capture modality for 3D Gaussian Splatting [Kerbl et al. 2023]: walk around a subject with a phone and let structure-from-motion [Schönberger and Frahm 2016] and differentiable rasterization do the rest. The first step — turning video into



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Table 1: Held-out novel-view quality and COLMAP registration rate per scene. Same frame budget, COLMAP settings, and training for both arms; only frame selection differs. Bold marks the better arm. Last row: paired mean delta with a cluster-bootstrap 95% CI (resampling scenes, then views).

Scene	PSNR $\uparrow$		SSIM $\uparrow$		LPIPS $\downarrow$		Reg. rate $\uparrow$	
	Unif.	Ours	Unif.	Ours	Unif.	Ours	Unif.	Ours
barn	25.68	24.62	0.801	0.784	0.272	0.286	100%	100%
chair	26.37	26.54	0.882	0.884	0.224	0.208	88%	95%
horse	23.34	22.62	0.873	0.866	0.175	0.180	100%	100%
ignatius	21.67	20.93	0.720	0.703	0.276	0.275	100%	100%
stone_dog	25.96	25.28	0.835	0.836	0.186	0.178	100%	100%
dl3dv_032dec9f	25.24	24.19	0.865	0.854	0.184	0.193	100%	100%
dl3dv_165f5af8	22.46	21.79	0.697	0.670	0.275	0.272	100%	100%
dl3dv_49381681	26.02	25.42	0.880	0.866	0.108	0.115	100%	100%
dl3dv_2beaca31	30.39	29.01	0.935	0.925	0.088	0.100	100%	100%
dl3dv_1ba74c22	29.04	28.79	0.932	0.933	0.229	0.238	100%	100%
Mean	25.62	24.92	0.842	0.832	0.202	0.205	99%	100%
Δ (95% CI)	-0.70 [-1.10, -0.33]		-0.010 [-0.017, -0.003]		+0.003 [-0.005, +0.010]			–

frames — is almost universally fixed-interval extraction (ffmpeg -r), widely believed to admit defects the pipeline cannot undo: *motion blur*, *redundancy*, and *uneven baselines*. We ask a narrow, practical question: *keeping the entire reconstruction pipeline fixed, how much does input curation alone buy?*

2 Method

Our tool (single Python file, OpenCV only) processes a video in three passes; every parameter is derived from the video itself. **(1) Analyze:** ~ 300 sampled points yield a sharpness distribution (Laplacian variance) and feature-track flow statistics, from which all thresholds are derived: the parallax trigger τ_{flow} (median tracked flow over the image diagonal, auto-adjusted to keep the estimated keyframe count in range), the blur floor $\tau_{\text{sharp}} = \min(P_{30}, 0.6 \cdot \text{median})$, a best-of buffer length, and a ~ 2 MP resolution cap. **(2) Select:** a keyframe fires when median tracked flow from the last keyframe exceeds τ_{flow} ; the sharpest buffered frame ($\geq \tau_{\text{sharp}}$) is kept. **(3) Configure:** the output is QC'd for parallax gaps and a ready-to-run COLMAP script is emitted (exhaustive, sequential, or loop-detection matching, chosen automatically).

3 Evaluation

Protocol. Each scene is one video. Test views (every 2 s, sharpest within ± 5 frames) are excluded from the selection pool of *both* arms. **Ours** runs the tool fully automatically; **Uniform** samples the *same number* of frames at equal intervals, same resolution cap, no blur handling — the standard ffmpeg recipe. Both arms share identical COLMAP settings (chosen by the tool's own QC), register the test frames jointly for poses, and train gsplat's standard trainer [Ye et al. 2025] for 15 k iterations at default hyperparameters, fixed seed. Scenes (10): Tanks and Temples source videos [Knapitsch et al. 2017] (*Barn*, *Horse*, *Ignatius*); five DL3DV captures [Ling et al. 2024], one per category chosen before inspecting outcomes; two casual phone captures (4K handheld *chair*, steady 1080p *stone_dog*).

Results. Table 1 summarizes; metrics are computed on the test views registered in *both* arms. On average curation shifts PSNR by -0.70 dB and SSIM by -0.010 (both bootstrap CIs exclude zero; LPIPS, DISTs, and FLIP are neutral-to-negative for curation). The

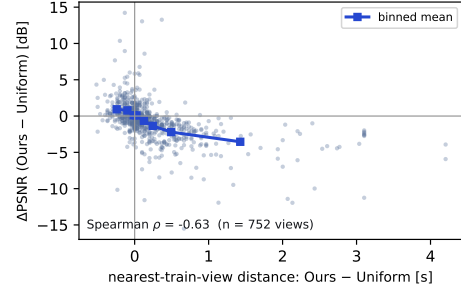


Figure 2: Per-view mechanism (752 jointly-registered test views): the PSNR difference tracks which arm has the temporally nearer training view.

only per-scene PSNR win (*chair*, $+0.17$ dB) lies within seed noise (two extra seeds move per-scene deltas by ± 0.2 – 0.4 dB); the mean deficit is far outside it. Notably, curation *does* deliver at the SfM stage — lower reprojection error, longer tracks, more 3D points, better registration (95% vs. 88% on the one incomplete scene) — yet this does not translate into novel-view quality.

Why: coverage predicts per-view quality. Mapping each test view to its temporal distance from the nearest training view in each arm explains the outcome (Fig. 2): the per-view PSNR difference correlates strongly with the *difference in local coverage* (Spearman $\rho = -0.63$; -0.58 to -0.74 in every scene). Where curation is locally denser it wins ($+0.69$ dB on average); where parallax-triggering leaves a sparser neighborhood it loses (-1.57 dB). Sharpness-based selection only pays when it does not cost coverage.

Controlled degradation study. To isolate input quality without scene confounds, we synthesize exposure blur from three 60 fps scenes by averaging $N \in \{1, 2, 4, 8\}$ consecutive frames in linear color space (GoPro protocol [Nah et al. 2017]); we name the levels blur-0/2/4/8 by the number of averaged frames, blur-0 meaning *no added blur* ($N=1$). Every level shares the scene, trajectory, lighting, codec, and the *same sharp* test views. Figure 3: the mean PSNR gap closes monotonically, $-0.93 \rightarrow -0.81 \rightarrow -0.70 \rightarrow -0.03$ dB ($+0.90$ dB recovered; one scene flips to $+0.90$). As degradation grows, blur rejection buys back what thinner coverage costs — but even 8-frame exposure only reaches parity on average.

4 Conclusion

At a matched frame budget, input curation does not improve novel-view quality on ordinary captures — despite genuinely improving SfM statistics — because per-view quality strongly tracks local temporal coverage, which parallax-triggered selection thins. Its deficit shrinks monotonically with input degradation ($+0.90$ dB recovered from blur-0 to blur-8, reaching near parity on average). Guidance: on ordinary captures prefer the plain uniform grid; under heavy degradation curation approaches parity on average and can help on individual scenes. Future selectors should preserve uniform coverage as a hard constraint. The tool and the full benchmark protocol are available at <https://github.com/UpHash-Network/curation-3dgs-benchmark>. Limitations: one representative selector (other

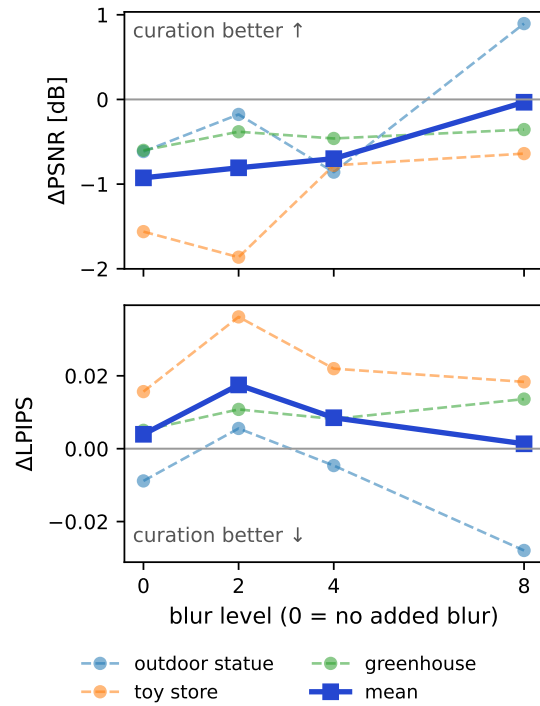


Figure 3: Synthetic-blur dose-response (3 scenes + mean): the mean PSNR gap (top) closes monotonically as simulated exposure grows; the LPIPS deltas (bottom) show the same trend but are not strictly monotonic.

strategies may differ); single-camera ordered video; ten real scenes, one trainer configuration; synthetic blur approximates real hand-held degradation.

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